

SAMIM REZA

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EDUCATION

Green University of Bangladesh <i>B.Sc. in Computer Science and Engineering</i>	2022 – 2025 <i>CGPA: 3.80/4.00</i>
Government Bangabandhu College <i>Higher Secondary Certificate</i>	2017 – 2019 <i>GPA: 4.75/5.00</i>
S. M. Model Govt. High School <i>Secondary School Certificate</i>	2012 – 2017 <i>GPA: 5.00/5.00</i>

WORK EXPERIENCE

Machine Learning Engineer — Brandifies	Feb 2025 – Present
– Built end-to-end ML pipelines: data preprocessing, feature engineering, model training, and evaluation – Fine-tuned LLMs and implemented RAG-based chatbot systems with vector databases – Developed predictive models using XGBoost and similar boosting algorithms	
Robotics Engineer Intern — RoboTech Valley	Jul – Sept 2025
– Designed robotics modules using embedded systems and ROS for autonomous navigation.	
Teaching Assistant — Green University of Bangladesh	May – Aug 2025
– Conducted demo classes on C Programming and Data Structures. – Evaluated exam scripts and prepared grade sheets.	
Programming Trainer — Green University of Bangladesh	Jan – Dec 2025
– Trained students for competitive programming through weekly problem-solving sessions.	
Mentor — University Mentorship Program	Jun 2023 – Aug 2025
– Mentored first-year students in programming fundamentals and professional tools.	

RESEARCH

Edge Detection for Smoke-Degraded Images using Fractional Derivative (Published)	2026
– Proposed a fractional calculus-based edge detection operator achieving robust boundary extraction under smoke and haze degradation conditions.	
Denoising Diffusion Probabilistic Models	
– Integrating edge detection for enhanced image generation.	
Federated Learning	
– Adaptive client selection algorithms for optimal convergence.	

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, PHP
AI/ML: PyTorch, TensorFlow, Scikit-learn, LangChain, VectorDB, RAG, LLM, Roboflow
Web/Backend: Django, FastAPI, React, REST APIs, PostgreSQL, MySQL, PWA
Tools & Platforms: ROS, Git, Docker, Celery, Redis, Meta Graph API, Linux, LaTeX, Render, Azure, Arduino, Raspberry Pi, Jetson, ESP32

FEATURED PROJECTS

- TrueDoc — Django, React, DRF, SimpleJWT, PostgreSQL, QR** GitHub
– SaaS for digital document attestation in Bangladesh with QR-backed verification, role-based access (citizen/verifier/org), JWT auth, and BDT payment model
- GoUp — Django, Celery, Redis, Meta Graph API, Twilio** GitHub
– For Lead Ads automation; syncs pages/forms via Meta API, webhooks, triggers SMS follow-ups Celery & Redis
- Personal AI Assistant — LLM, RAG, LangChain** Live | GitHub
– Built an LLM-powered chatbot with retrieval-augmented generation for personalized assistance.
- KobitaLM — Bengali Poetry Generator** Model | GitHub
– Fine-tuned Gemma 2 9B with LoRA and RAG for Bengali poetry generation.
- WeCare Medical Service App — FastAPI, Qwen3-VL, PWA** GitHub
– Developed an offline-first PWA for rural healthcare with AI-assisted triage capabilities.
- Cirrhosis Outcome Prediction — XGBoost, LightGBM, CatBoost** BAIC 2025
– Built an ensemble ML classifier with 55 engineered features, achieving 14.5% accuracy gain.
- Hasib Academic System — Django, PostgreSQL, Render, Gunicorn** Live | GitHub
– A production-ready Django application for school workflows including admissions, student attendance, notices, teacher accounts, and finance tracking with PDF export. Tailored for Bangla-speaking users.
- Union Ledger — Django, PostgreSQL, Render** Live | GitHub
– Built a collaborative financial tracking web app with consensus-based voting for deposits, loans, repayments, and investments.
- Mars Rover — ROS, Embedded Systems** GitHub
– Developed a ROS-based autonomous navigation system for Mars rover simulation.

COMPETITIVE PROGRAMMING

- Champion: Intra Dept Programming Contest 2024.
- 3x ICPC Regionalist: 2022, 2023, 2024.
- Around 1000 problems solved across online judges.
- Kaggle Competitor: Active in ML competitions.
- Problem Setter: Junior Programming Contests.
- Participated: NCPC 2023, IUPC 2023/2024.

ACHIEVEMENTS

- Vice Chancellor's and Dean's Award for Academic Excellence.
- Champion, Inter Dept Programming Contest 2024.
- Secured 8th position in BUBT Artificial Intelligence Conquest (Kaggle).
- National IT Quiz Competition Medalist.

REFERENCES

Prof. Dr. Md. Saiful Azad
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